

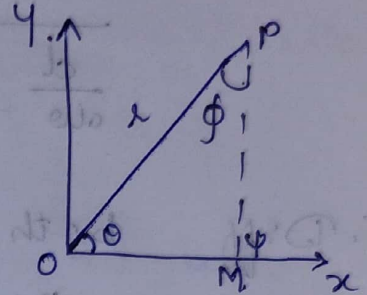
$$\underline{\underline{M-1}}$$

1. Angle b/w radius vector & tangent

$$\tan \phi = r \frac{dy}{dx} \quad \text{or} \quad \cot \phi = \frac{1}{r} \frac{dx}{dy}$$

$\Rightarrow$ Consider a point P on the curve, $r = f(\theta)$.

Let, OP be radius vector r
 PM be tangent.



θ is angle b/w radius vector & x -axis

ϕ is angle b/w radius vector & tangent

ψ is angle b/w radius vector & x -axis.

WKT, Outer angle is equal to sum of interior angles.

$$\therefore \psi = \phi + \theta$$

taking tan on both sides.

$$\tan \psi = \tan(\phi + \theta)$$

$$\tan \psi = \tan(\phi + \theta)$$

$$\tan \psi = \frac{\tan \phi + \tan \theta}{1 - \tan \phi \cdot \tan \theta}$$

$$1 - \tan \phi \cdot \tan \theta \quad \text{--- ①}$$

Since, $x = r \cos \theta$ & $y = r \sin \theta$.

$$\tan \psi = \frac{dy}{dx}$$

$$\tan \psi = \frac{dy}{dx} \frac{dx}{d\theta}$$

$$\tan \psi = \frac{\frac{d}{d\theta} (r \sin \theta)}{\frac{d}{d\theta} (r \cos \theta)}$$

∴ Diff both Nr and dr by using product rule.

$$\tan \psi = \frac{r \cos \theta + \sin \theta (r')}{r(-\sin \theta) + \cos \theta (r')}$$

$$\tan \psi = \frac{r(\cos \theta) + \sin \theta (r')}{-r(\sin \theta) + \cos \theta (r')}$$

÷ by $\cos \theta (r')$.

$$\tan \psi = \frac{\cancel{r \cos \theta}}{\cancel{\cos \theta} r'} + \frac{\sin \theta r}{\cos \theta r}$$

$$= \frac{-r \sin \theta}{\cos \theta r} + \frac{\cancel{\cos \theta} r}{\cancel{\cos \theta} r}$$

$$= \frac{r}{r'} + \tan \theta$$

$$= \frac{1}{\tan \theta} + 1$$

$$= \frac{1}{\tan \theta} + \tan \theta$$

Comparing eq ① & ②,

$$\tan \psi = \frac{r}{r'}$$

$$= \frac{r}{\frac{dr}{d\theta}}$$

$$\tan \phi = r \frac{d\theta}{dr}$$

②

$$\cot \phi = \frac{1}{r} \frac{dr}{d\theta}$$

$$\frac{r}{r'} = \frac{10}{90} = \frac{1}{9}$$

$$\phi = 2 \text{ or } 9$$



Q2) prove pedal equation.

$$\frac{1}{p^2} = \frac{1}{r^2} + \frac{1}{r^4} \left(\frac{dr}{d\theta} \right)^2$$

Consider the point P on the curve, $r = f(\theta)$.

Draw OP radius vector denoted as r
Draw the tangent at the point

Draw OL perpendicular from pole to the tangent.

Let $\angle OLP = 90^\circ$

θ is the angle b/w radius vector and x-axis.

ϕ is the angle b/w radius vector and tangent.

from right angle triangle $\triangle OLP$,

$$\sin \phi = \frac{OL}{OP} = \frac{p}{r}$$

$$p = r \sin \phi$$

Taking reciprocal $\frac{1}{p} = \frac{1}{r \sin \phi}$

Squaring O.B.S. $(\sin^2 \phi + 1) r^2 = r^2$

$$\frac{1}{p^2} = \frac{1}{r^2 \sin^2 \phi}$$

$$\frac{1}{p} = \frac{1}{r^2} \cdot \sec^2 \phi$$

$$= \frac{1}{r^2} \cdot (1 + \cot^2 \phi)$$

$$= \frac{1}{r^2} \cdot \left(1 + \left(\frac{1}{r} \frac{dr}{d\theta} \right)^2 \right)$$

$$\frac{1}{p} = \frac{1}{r^2} + \frac{1}{r^2} \left(\frac{dr}{d\theta} \right)^2$$

$$\frac{1}{p^2} = \frac{1}{r^2} + \frac{1}{r^2} \left(\frac{dr}{d\theta} \right)^2$$

03. Find angle b/w radius vector & tangent.

i) $r = a(1 + \cos \theta)$

→ Consider,

$$r = a(1 + \cos \theta)$$

taking log o.B.S.

$$\log r = \log [a(1 + \cos \theta)]$$

$$\log r = \log a + \log(1 + \cos \theta)$$

Diff w.r.t θ

$$\frac{1}{r} \frac{dr}{d\theta} = 0 + \frac{1}{1 + \cos \theta} (0 + (-\sin \theta))$$

$$\frac{1}{r} \frac{dr}{d\theta} = \frac{-\sin \theta}{1 + \cos \theta}$$

$$\cot \phi = \frac{-\sin \theta / 2 \cdot \cos \theta / 2}{2 \cdot \cos^2 \theta / 2}$$

$$= \frac{-\sin \theta / 2}{\cos \theta / 2}$$

$$\cot \phi = -\tan \frac{\theta}{2}$$

$$\cot \phi = \cot \left(\frac{\pi}{2} + \frac{\theta}{2} \right)$$

$$\boxed{\phi = \frac{\pi}{2} + \frac{\theta}{2}}$$

ii) $r = a(1 - \cos \theta)$

taking log o.B.S.

$$\log r = \log [a(1 - \cos \theta)]$$

$$\log r = \log a + \log(1 - \cos \theta)$$

Diff w.r.t θ

$$\frac{1}{r} \frac{dr}{d\theta} = 0 + \frac{1}{1 - \cos \theta} (0 - (-\sin \theta))$$

$$\frac{1}{r} \frac{dr}{d\theta} = \frac{\sin \theta}{1 - \cos \theta}$$

$$\cot \phi = \frac{2 \sin \theta / 2 \cdot \cos \theta / 2}{2 \sin^2 \theta / 2}$$

$$= \frac{\sin \theta / 2 \cdot \cos \theta / 2}{\sin^2 \theta / 2}$$

$$\cot \phi = \cot \theta / 2$$

$$\boxed{\phi = \frac{\theta}{2}}$$

$$\text{iii) } r^m = a^m (\cos m\theta + \sin m\theta)$$

taking log obs.

$$\log r^m = \log [a^m (\cos m\theta + \sin m\theta)]$$

$$= \log a^m + \log (\cos m\theta + \sin m\theta)$$

$$\log r^m = \log a^m + \log (\cos m\theta + \sin m\theta)$$

$$m \log r = m \log a + \log (\cos m\theta + \sin m\theta)$$

diff w.r.t θ .

$$m \cdot \frac{1}{r} \frac{dr}{d\theta} = 0 + \frac{1}{(\cos m\theta + \sin m\theta)} (-\sin m\theta + \cos m\theta)$$

$$m \cot \phi = \frac{(-\sin m\theta + \cos m\theta)}{\cos m\theta + \sin m\theta}$$

$$m \cot \phi = \frac{(\cos m\theta - \sin m\theta)}{\cos m\theta + \sin m\theta}$$

taking N_1 & D_1 common.

$$\cot \phi = \frac{\cos m\theta \left(1 - \frac{\sin m\theta}{\cos m\theta}\right)}{\cos m\theta \left(1 + \frac{\sin m\theta}{\cos m\theta}\right)}$$

$$\cot \phi = \frac{1 - \tan m\theta}{1 + \tan m\theta}$$

$$\cot \phi = \cot \left(\frac{\pi}{4} + m\theta \right)$$

$$\boxed{\phi = \frac{\pi}{4} + m\theta}$$

Q4. Show that points of polar curve intersecting orthogonally.

i) $r = a(1 + \cos \theta)$ & $r = b(1 - \cos \theta)$.

Consider, $r = a(1 + \cos \theta)$.

taking log obs.

$$\log r = \log [a(1 + \cos \theta)]$$

$$\log r = \log a + \log (1 + \cos \theta)$$

diff w.r.t θ .

$$\frac{1}{r} \frac{dr}{d\theta} = 0 + \frac{1}{1 + \cos \theta} (0 + (-\sin \theta))$$

$$\frac{1}{r} \cdot \frac{dr}{d\theta} = \frac{-\sin \theta / 2}{1 + \cos \theta}$$

$$\cot \phi_1 = \frac{-2 \sin \theta / 2 \cdot \cos \theta / 2}{2 \cos^2 \theta / 2}$$

$$= \frac{-\sin \theta / 2}{\cos \theta / 2}$$

$$\cot \phi_1 = -\tan \theta / 2$$

$$\cot \phi_1 = \cot \left(\frac{\pi}{2} + \frac{\theta}{2} \right)$$

$$\boxed{\phi_1 = \frac{\pi}{2} + \frac{\theta}{2}}$$

• Consider, $r = b(1 - \cos \theta)$.

taking log OBS.

$$\log r = \log [b(1 - \cos \theta)]$$

$$\log r = \log b + \log (1 - \cos \theta)$$

$$\frac{1}{r} \frac{dr}{d\theta} = 0 + \frac{1}{1 - \cos \theta} (0 - (-\sin \theta))$$

$$\frac{1}{r} \frac{dr}{d\theta} = \frac{\sin \theta}{1 - \cos \theta}$$

$$\cot \phi_2 = \frac{2 \sin \theta/2 \cdot \cos \theta/2}{2 \sin^2 \theta/2}$$

$$\cot \phi_2 = \frac{\cos \theta/2}{\sin \theta/2}$$

$$\cot \phi_2 = \cot \theta/2$$

$$\phi_2 = \theta/2 \quad \therefore \text{it is orthogonal.}$$

• Angle of intersection = $|\phi_1 - \phi_2|$

$$\left(\frac{\theta}{2} + \frac{\pi}{2}\right) - \left(\frac{\pi}{2} + \frac{\theta}{2} - \frac{\theta}{2}\right)$$

$$\frac{\theta}{2} + \frac{\pi}{2} - \frac{\pi}{2}$$

ii) $r^n = a^n \cos n\theta$ & $r^n = b^n \sin n\theta$.

• Consider, $r^n = a^n \cos n\theta$.

taking log OBS.

$$\log r^n = \log [a^n (\cos n\theta)]$$

$$\log r^n = \log a^n + \log \cos n\theta$$

$$n \log r = n \log a + \frac{\log \cos n\theta}{\cos n\theta} (-\sin n\theta)(n)$$

diff w.r.t θ .

$$n \frac{1}{r} \frac{dr}{d\theta} = 0 + \frac{1}{\cos n\theta} (-\sin n\theta) n$$

$$\frac{1}{r} \frac{dr}{d\theta} = -\frac{\sin n\theta}{\cos n\theta}$$

$$\cot \phi_1 = -\tan n\theta$$

$$\cot \phi_1 = \cot \left(\frac{\pi}{2} + n\theta\right)$$

$$\phi_1 = \left(\frac{\pi}{2} + n\theta\right)$$

• Consider, $r^n = b^n \sin n\theta$.

taking log OBS.

$$\log r^n = \log [b^n (\sin n\theta)]$$

$$\log r^n = \log b^n + \log \sin n\theta$$

$$n \log r = n \log b + \log \sin n\theta$$

Diff w.r.t θ .

$$r \cdot \frac{1}{r} \frac{dr}{d\theta} = 0 + \frac{1}{\sin n\theta} (\cos n\theta)$$

$$\frac{1}{r} \frac{dr}{d\theta} = \frac{\cos n\theta}{\sin n\theta}$$

$$\cot \phi_2 = \cot n\theta$$

$$\phi_2 = n\theta$$

$$\text{Angle of intersection} = |\phi_1 - \phi_2|$$

$$= \left| \frac{\pi}{2} + n\theta - n\theta \right|$$

$$= \frac{\pi}{2}$$

$$\left(\theta + \frac{\pi}{2} \right) \text{ to } \theta = \frac{\pi}{2}$$

$$\left(\theta + \frac{\pi}{2} \right) = \frac{\pi}{2}$$

$$\text{iii) } r^2 \sin 2\theta = 4 \quad \& \quad r^2 = 16 \sin 2\theta$$

find the pair of curve.

$$\text{Consider, } r^2 \sin 2\theta = 4$$

$$\text{taking log obs.}$$

$$\log r^2 \sin 2\theta = \log 4$$

$$\log r^2 + \log \sin 2\theta = \log 4$$

$$2 \log r + \log \sin 2\theta = \log 4$$

$$\text{diff w.r.t } \theta$$

$$2 \cdot \frac{1}{r} \frac{dr}{d\theta} + \frac{1}{\sin 2\theta} (\cos 2\theta)(2) = 0$$

$$2 \cdot \cot \phi_1 + \frac{\cos 2\theta}{\sin 2\theta} (2) = 0$$

$$2 \cot \phi_1 + \cot 2\theta (2) = 0$$

$$2 \cot \phi_1 = -2 \cot 2\theta$$

$$\phi_1 = -2\theta$$

$$\text{Consider, } r^2 = 16 \sin 2\theta$$

$$\text{taking log obs.}$$

$$\log r^2 = \log (16 \sin 2\theta)$$

$$\log r^2 = \log 16 + \log \sin 2\theta$$

$$2 \log r = \log 16 + \log \sin 2\theta$$

$$\text{diff w.r.t } \theta$$

$$2 \cdot \frac{1}{r} \frac{dr}{d\theta} = 0 + \frac{1}{\sin 2\theta} (\cos 2\theta)(2)$$

$$2 \cdot \frac{1}{k} \frac{du}{d\theta} = \frac{\cos 2\theta}{\sin 2\theta} \quad (2)$$

$$2 \cot \phi_2 = 2 \cot 2\theta$$

$$\boxed{\phi_2 = 2\theta}$$

• Angle of intersection: $|\phi_1 - \phi_2|$
 $= |-2\theta + 2\theta|$
 $= |0|$
 $= 0 \quad \text{--- ①}$

• To solve for θ ;

Consider given equation:

$$r^2 \sin 2\theta = 4 \quad \text{and} \quad r^2 = 16 \sin 2\theta$$

$$r^2 = \frac{4}{\sin 2\theta} \quad r^2 = 16 \sin 2\theta$$

$$\therefore \frac{4}{\sin 2\theta} = 16 \sin 2\theta$$

$$\frac{1}{\sin 2\theta} = 4 \sin 2\theta$$

$$1 = 4 \sin^2 2\theta$$

$$\sin^2 2\theta = \frac{1}{4}$$

$$\sin 2\theta = \sqrt{\frac{1}{4}}$$

$$\sin 2\theta = \frac{1}{2}$$

$$2\theta = \sin^{-1}\left(\frac{1}{2}\right)$$

$$\theta = \frac{\pi}{6}$$

$$\boxed{\theta = \frac{\pi}{12}}$$

① $\rightarrow$

$$|\phi_1 - \phi_2| = 4\theta$$

$$= 4 \left(\frac{\pi}{12} \right)$$

$$= \frac{\pi}{3}$$

$\therefore$ it is not orthogonal

$$iv) x = a \log \theta. \quad \& \quad x = \frac{a}{\log \theta}.$$

• Consider, $x = a \log \theta$.

taking log o.B.S.

$$\log x = \log (a \log \theta)$$

$$\log x = \log a + \log (\log \theta).$$

$$\frac{1}{x} \frac{dx}{d\theta} = 0 + \frac{1}{\log \theta} \left(\frac{1}{\theta} \right).$$

$$\cot \phi_1 = \frac{1}{\log \theta} \times \left(\frac{1}{\theta} \right).$$

$$\tan \phi_2 = \frac{0 \log \theta}{1}.$$

• Consider, $x = \frac{a}{\log \theta}$.

taking o.B.S.

$$\log x = \log \left(\frac{a}{\log \theta} \right).$$

$$\log x = \log a - \log \theta (\log \theta).$$

diff wrt θ

$$\frac{1}{x} \frac{dx}{d\theta} = 0 - \frac{1}{\log \theta} \left(\frac{1}{\theta} \right).$$

$$\cot \phi_2 = \frac{1}{\log \theta} \times \frac{1}{\theta}.$$

$$\tan \phi_2 = \frac{-0 (\log \theta)}{1}.$$

• Consider, $\tan (\phi_1 - \phi_2)$.

$$\begin{aligned} \tan (\phi_1 - \phi_2) &= \frac{\tan \phi_1 - \tan \phi_2}{1 + \tan \phi_1 \cdot \tan \phi_2} \\ &= \frac{0 (\log \theta) - (-0 (\log \theta))}{1 + 0 (\log \theta) \cdot (0 (\log \theta))} \end{aligned}$$

Consider equation.

$$x = a \log \theta.$$

$$x = \frac{a}{\log \theta}.$$

$$a \log \theta = \frac{a}{\log \theta}$$

$$\log^2 \theta = 1$$

$$\log \theta = 1.$$

• Substitute 1 in eq ①.

$$\begin{aligned} \tan (\phi_1 - \phi_2) &= \frac{0 (\log \theta) - (-0 \cdot \log \theta)}{1 + 0 \log \theta \cdot (-0 \cdot \log \theta)} \\ &= \frac{0 (1) - 0 (1)}{1 + 0 (1) - 0 (1)} \\ &= \frac{20}{1 - 0^2}. \end{aligned}$$

$$\tan (\phi_1 - \phi_2) = \frac{20}{1 - 0^2}$$

$$\boxed{(\phi_1 - \phi_2) = \tan^{-1} \left(\frac{20}{1 - 0^2} \right)}$$

$$\text{if } r = a(1 + \cos \theta).$$

taking log obs.

$$\log r = \log [a(1 + \cos \theta)]$$

$$\log r = \log a + \log (1 + \cos \theta).$$

diff wrt θ .

$$\frac{1}{r} \frac{dr}{d\theta} = 0 + \frac{1}{1 + \cos \theta} (0 + (-\sin \theta))$$

$$\frac{1}{r} \frac{dr}{d\theta} = -\frac{\sin \theta}{1 + \cos \theta}.$$

$$\frac{1}{r} \frac{dr}{d\theta} = \frac{-\sin \theta/2 \cdot \cancel{\cos \theta/2}}{\cancel{2} \cos^2 \theta/2}.$$

$$\cot \phi = \frac{-\sin \theta/2}{\cos \theta/2}.$$

$$\cot \phi = -\tan \theta/2.$$

$$\cot \phi = \cot \left(\frac{\pi}{2} + \frac{\theta}{2} \right)$$

$$\boxed{\phi = \frac{\pi}{2} + \frac{\theta}{2}}$$

Consider pedal eqⁿ:

$$p = r \sin \phi$$

$$p = r \sin \left(\frac{\pi}{2} + \frac{\theta}{2} \right)$$

$$\underline{p = r \cos \frac{\theta}{2}} \quad \text{--- (A)}$$

Consider given eqⁿ:

$$r = a(1 + \cos \theta).$$

$$r = a \left(a \cos^2 \frac{\theta}{2} \right)$$

$$\cos \frac{2\theta}{2} = \frac{r}{2a}$$

$$\cos \frac{\theta}{2} = \sqrt{\frac{r}{2a}}$$

Substitute $\cos \frac{\theta}{2}$ in eq (A)

$$p = r \cos \frac{\theta}{2}$$

$$p = r \cdot \sqrt{\frac{r}{2a}}$$

$$p = r \cdot \frac{\sqrt{r}}{\sqrt{2a}}$$

$$= \frac{r^1 r^{1/2}}{\sqrt{2a}}$$

$$= \frac{r^{1+1/2}}{\sqrt{2a}}$$

$$= r^{2+1/2} / \sqrt{2a}$$

$$\boxed{p = \frac{r^{3/2}}{\sqrt{2a}}}$$



iii) $r = ae^{\theta \cot \alpha}$

Consider,
 $r = ae^{\theta \cot \alpha}$

taking log obs.

$$\log r = \log ae^{\theta \cot \alpha}$$

$$\log r = \log a + \log e^{\theta \cot \alpha}$$

$$\log r = \log a + \theta \cot \alpha \log e$$

diff w.r.t θ .

$$\frac{1}{r} \frac{dr}{d\theta} = 0 + (1) \cot \alpha (1) \cdot 1 = \cot \alpha$$

$$\cot \phi = \cot \alpha$$

$$\boxed{\phi = \alpha}$$

Consider pedal eqn

$$p = r \sin \phi$$

$$p = r \sin \alpha$$

diff w.r.t p .

$$1 = \sin \alpha \frac{dr}{dp}$$

$$\therefore \frac{dr}{dp} = \frac{1}{\sin \alpha}$$

$$\boxed{\frac{dr}{dp} = \operatorname{cosec} \alpha}$$

Consider Radius of curvature

$$\rho = r \frac{dr}{dp}$$

$$\rho = r \frac{1}{\sin \alpha}$$

$$\frac{\rho}{r} = \frac{1}{\sin \alpha}$$

$$\frac{\rho}{r} = \operatorname{cosec} \alpha$$

$$\underline{\underline{\frac{\rho}{r} = \text{Constant}}}$$

06. Find the radius of curvature

$$\rho = \frac{(1 + y_1^2)^{3/2}}{y_2}$$

Consider the curve $y = f(x)$

$$\tan \psi = \frac{dy}{dx} = y_1$$

$$\tan \psi = y_1$$

$$\psi = \tan^{-1} y_1$$

diff w.r.t x .

$$\frac{d\psi}{dx} = \frac{1}{1 + (y_1^2)} \cdot (y_2) \quad \left| \frac{d}{dx} \tan^{-1}(x) = \frac{1}{1+x^2} \right.$$

$$\frac{d\psi}{dx} = \frac{y_2}{1 + y_1^2}$$

Consider arc length. $\frac{ds}{dx} = \sqrt{1 + y_1^2}$

Radius of curvature, $\rho = \frac{ds}{d\psi}$

$$\rho = \frac{\frac{ds}{dx}}{\frac{d\psi}{dx}}$$

$$\rho = \frac{\sqrt{1 + y_1^2}}{\frac{y_2}{1 + y_1^2}}$$

$$\rho = \frac{\sqrt{1 + y_1^2} \times 1 + y_1^2}{y_2}$$

$$= \frac{(1 + y_1^2)^{1/2} (1 + y_1^2)}{y_2}$$

$$\rho = \frac{(1 + y_1^2)^{3/2}}{y_2}$$



07. Find the radius of Curvature of

if $y = a \log \left(\sec \frac{x}{a} \right)$

Consider,

$$y = a \log \sec \left(\frac{x}{a} \right)$$

diff wrt x

$$\therefore \frac{dy}{dx} = y_1$$

$$y_1 = a \cdot \frac{1}{\sec \frac{x}{a}} \left(\sec \frac{x}{a} \right) \cdot \tan \left(\frac{x}{a} \right) \cdot \frac{1}{a} \quad (1)$$

$$y_1 = \tan \frac{x}{a}$$

Once again diff wrt x .

$$y_2 = \frac{1}{a} \sec^2 \frac{x}{a}$$

Radius of curvature.

$$\rho = \frac{(1 + y_1^2)^{3/2}}{y_2}$$

$$\rho = \frac{\left(1 + \left(\tan \frac{x}{a} \right)^2 \right)^{3/2}}{\frac{1}{a} \sec^2 \frac{x}{a}}$$

$$\rho = \frac{\left(1 + \tan^2 \frac{x}{a} \right)^{3/2}}{\frac{1}{a} \sec^2 \frac{x}{a}}$$

$$\rho = \frac{\left(\sec^2 \frac{x}{a} \right)^{3/2}}{\frac{1}{a} \sec^2 \frac{x}{a}}$$

$$\rho = \frac{\sec^3 \frac{x}{a}}{\frac{1}{a} \sec^2 \frac{x}{a}}$$

$$\boxed{\rho = a \sec \frac{x}{a}}$$



iii) $x^3 + y^3 = 3axy$ at $\left(\frac{3a}{2}, \frac{3a}{2}\right)$.

Consider $x^3 + y^3 = 3axy$.

Diff wrt x .

$$3x^2 + 3y^2 \cdot y_1 = 3a[x \cdot y_1 + y(1)]$$

$$3x^2 + 3y^2 \cdot y_1 = 3axy_1 + 3ay$$

$$3y^2 y_1 - 3axy_1 = 3ay - 3x^2$$

$$(3y^2 - 3ax) = 3ay - 3x^2$$

$$y_1 = \frac{3ay - 3x^2}{3y^2 - 3ax}$$

Given pt. $\left(\frac{3a}{2}, \frac{3a}{2}\right)$.

$$y_1 = \frac{3a\left(\frac{3a}{2}\right) - 3\left(\frac{3a}{2}\right)^2}{3\left(\frac{3a}{2}\right)^2 - 3a\left(\frac{3a}{2}\right)}$$

$$= \frac{-\left(3\left(\frac{3a}{2}\right)^2 - 3a\left(\frac{3a}{2}\right)\right)}{3\left(\frac{3a}{2}\right)^2 - 3a\left(\frac{3a}{2}\right)}$$

$$\boxed{y_1 = -1}$$

Since,

$$y_1 = \frac{3ay - 3x^2}{3y^2 - 3ax}$$

$$y_1(3y^2 - 3ax) = 3ay - 3x^2$$

$$y_1(3y^2 \cdot y_1 - 3a(1)) + (3y^2 - 3ax) = 3ay_1 - 3x^2$$

$$6y^2 y_1 - 3ay_1 + (3y^2 - 3ax)y_2 = 3ay_1 - 6x$$

$$(3y^2 - 3ax)y_2 = 3ay_1 - 6x - 6y^2 y_1 + 3ay_1$$

$$y_2 = \frac{3ay_1 - 6x - 6y^2 y_1 + 3ay_1}{3y^2 - 3ax}$$

$$y_2 = \frac{3a(-1) - 6\left(\frac{3a}{2}\right) - 6\left(\frac{3a}{2}\right)(-1)^2 + 3a(-1)}{3\left(\frac{3a}{2}\right)^2 - 3a\left(\frac{3a}{2}\right)}$$

$$= \frac{-3a - 9a - 9a - 3a}{\frac{27a^2}{4} - \frac{9a^2}{2}}$$

$$= \frac{24a}{27a^2 - 18a^2}$$

$$= \frac{24a}{-9a^2} \Rightarrow \boxed{y_2 = -\frac{32}{3a}}$$

• Radius of curvature:

$$r = \frac{(1 + y_1^2)^{3/2}}{y_2}$$

$$= \frac{(1 + (-1)^2)^{3/2}}{-\frac{32}{3a}}$$

$$= \frac{(2)^{3/2}}{-\frac{32}{3a}}$$

$$= 2\sqrt{2} \times \frac{-3a}{32}$$

$$r = -\frac{3a\sqrt{2}}{16}$$



$$\rho = \frac{-3a\sqrt{2}}{8 \times 2}$$

$$\rho = \frac{-3a\sqrt{2}}{8 \times \sqrt{2} \cdot \sqrt{2}}$$

$$\rho = \frac{-3a}{8 \times \sqrt{2}}$$

∴ Hence ρ is always +ve, $\rho = \frac{3a}{8 \times \sqrt{2}}$

$$\text{iii) } y = c \cosh\left(\frac{x}{c}\right)$$

Consider,

$$y = c \cosh\left(\frac{x}{c}\right)$$

diff w.r.t x

$$y_1 = c \sinh\left(\frac{x}{c}\right) \cdot \frac{1}{c}$$

$$\boxed{y_1 = \sinh\left(\frac{x}{c}\right)}$$

Once again diff w.r.t x

$$y_2 = \cosh\left(\frac{x}{c}\right) \cdot \frac{1}{c}$$

$$\boxed{y_2 = \frac{1}{c} \cosh\left(\frac{x}{c}\right)}$$

Radius of curvature:

$$\rho = \frac{(1 + y_1^2)^{3/2}}{y_2}$$

$$\rho = \left[1 + \sinh^2\left(\frac{x}{c}\right)\right]^{3/2}$$

$$\rho = \frac{1}{c} \cosh\left(\frac{x}{c}\right)$$

$$= \frac{\left[1 + \sinh^2\left(\frac{x}{c}\right)\right]^{3/2}}{\frac{1}{c} \cosh\left(\frac{x}{c}\right)}$$

$$= \frac{\left[\cosh^2\left(\frac{x}{c}\right)\right]^{3/2}}{\frac{1}{c} \cosh\left(\frac{x}{c}\right)}$$

$$\rho = c \cosh^2\left(\frac{x}{c}\right)$$

Consider given eqn:

$$y = c \cosh\left(\frac{x}{c}\right)$$

$$\cosh\left(\frac{x}{c}\right) = \frac{y}{c}$$

$$\cosh^2\left(\frac{x}{c}\right) = \frac{y^2}{c^2}$$

Substitute in ①.

$$\rho = c \cosh^2\left(\frac{x}{c}\right)$$

$$\rho = c \left(\frac{y^2}{c^2}\right)$$

$$\boxed{\rho = \frac{y^2}{c}}$$



iv) Find ρ of Cycloid $x = a(\theta + \sin \theta)$,
 $y = a(1 - \cos \theta)$.

Consider,

$$x = a(\theta + \sin \theta)$$

diff wrt x

$$\frac{dx}{d\theta} = a(1 + \cos \theta)$$

Consider,

$$y = a(1 - \cos \theta)$$

diff wrt y .

$$\frac{dy}{d\theta} = a(0 - (-\sin \theta))$$

$$\frac{dy}{d\theta} = a \sin \theta$$

Consider,

$$y_1 = \frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$$

$$= \frac{a \sin \theta}{a(1 + \cos \theta)}$$

$$= \frac{\cancel{a} \cdot \cancel{2} \sin \theta / 2 \cdot \cos \theta / 2}{\cancel{a} \cdot \cancel{2} \cos^2 \theta / 2}$$

$$= \frac{\sin \theta / 2}{\cos \theta / 2}$$

$$y_1 = \tan \theta / 2$$

diff again.

$$y_2 = \sec^2 \theta / 2 \left(\frac{1}{2} \right) \left(\frac{d\theta}{dx} \right)$$

$$= \sec^2 \frac{\theta}{2} \left(\frac{1}{2} \right) \left(\frac{1}{a(1 + \cos \theta)} \right)$$

$$= \sec^2 \frac{\theta}{2} \cdot \frac{1}{2} \cdot \frac{1}{a(2 \cos^2 \frac{\theta}{2})}$$

$$= \frac{1}{4a} \cdot \sec^2 \frac{\theta}{2} \cdot \sec^2 \frac{\theta}{2}$$

$$y_2 = \frac{1}{4a} \cdot \sec^4 \frac{\theta}{2}$$

Radius of curvature:

$$\rho = \frac{(1 + y_1^2)^{3/2}}{y_2}$$

$$= \frac{(1 + (\tan \theta / 2)^2)^{3/2}}{\frac{1}{4a} \cdot \sec^4 \frac{\theta}{2}}$$

$$= \frac{(1 + \tan^2 \frac{\theta}{2})^{3/2}}{\frac{1}{4a} \cdot \sec^4 \frac{\theta}{2}}$$

$$= \frac{(1 + \tan^2 \frac{\theta}{2})^{3/2}}{\frac{1}{4a} \cdot \sec^4 \frac{\theta}{2}}$$

$$= \frac{1}{4a} \cdot \frac{(1 + \tan^2 \frac{\theta}{2})^{3/2}}{\sec^4 \frac{\theta}{2}}$$

$$= \frac{1}{4a} \cdot \frac{(1 + \tan^2 \frac{\theta}{2})^{3/2}}{\sec^4 \frac{\theta}{2}}$$

$$\boxed{\rho = 4a \cdot \cos \frac{\theta}{2}}$$

v) $r^n = a^n \cos n\theta$. & prove that p is inversely $\propto$ to r^{n-1} .

Consider, $r^n = a^n \cos n\theta$.

taking log on both sides.

$$\log r^n = \log [a^n \cos n\theta]$$

$$\log r^n = \log a^n + \log \cos n\theta$$

$$n \cdot \log r = n \log a + \log \cos n\theta$$

diff w.r.t θ .

$$n \cdot \frac{1}{r} \frac{dr}{d\theta} = 0 + \frac{1}{\cos n\theta} (-\sin n\theta)(n)$$

$$\frac{1}{r} \frac{dr}{d\theta} = \frac{-\sin n\theta}{\cos n\theta} = -\tan n\theta$$

$$\cot \phi = -\tan n\theta$$

$$\cot \phi = \cot \left[\frac{\pi}{2} + n\theta \right]$$

$$\boxed{\phi = \left(\frac{\pi}{2} + n\theta \right)}$$

Consider pedal equation:

$$p = r \sin \phi$$

$$p = r \sin \left(\frac{\pi}{2} + n\theta \right)$$

$$p = r \cos n\theta \quad \text{--- (1)}$$

Consider given eqn:

$$r^n = a^n \cos n\theta$$

$$\boxed{\cos n\theta = \frac{r^n}{a^n}}$$

Sub $\frac{r^n}{a^n}$ in eq (1) $p = \frac{r^n}{a^n}$

$$p = r \cos n\theta$$

$$= r \frac{r^n}{a^n}$$

$$p = \frac{r^{1+n}}{a^n}$$

Diff w.r.t p .

$$1 = \frac{1}{a^n} (1+n) r^{1+n-1} \cdot \frac{dr}{dp}$$

$$1 = \frac{1+n}{a^n} \cdot r^n \cdot \frac{dr}{dp}$$

$$\frac{dr}{dp} = \frac{a^n}{(1+n) \cdot r^n}$$

Consider radius of Curvature:

$$\rho = r \frac{dr}{dp}$$

$$\rho = r \cdot \frac{a^n}{(1+n) r^n}$$

$$= \frac{a^n}{(1+n) r^{n-1}}$$

$$\rho = \frac{a^n}{(1+n) r^{n-1}}$$

$\therefore \rho$ is inversely proportional to given equation.